DEGREE FLUCTUATIONS IN BILATERAL PREFERENCE RANDOM GRAPHS

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Abstract. In the bilateral preference random graph $LK(n,k)$ of La and Kabkab, independent uniform scores are placed on the edges of K_n, each vertex prefers its k incident edges of smallest score, and an edge is kept when both of its endpoints prefer it. Dabirian and Subramanian asked for the limiting distribution of the standardized degree $(D-\mathbb{E}D)/\sqrt{\operatorname{Var} D}$ of a vertex. We show that if $k\to\infty$ and $n-k\to\infty$, then $(k-D)/s_n$, where $s_n^2=k(n-k)/n$, converges in distribution and in its first two moments to $\phi(Z)+Z\Phi(Z)$ with Z standard normal. Consequently $\mathbb{E}D=k-s_n/\sqrt{\pi}+o(s_n)$, $\operatorname{Var} D=(1/3+(\sqrt{3}-2)/(2\pi))s_n^2+o(s_n^2)$, and the standardized degree converges to an explicit law that is bounded above, hence not normal. If instead $r=n-1-k\geq 1$ is fixed, then $k-D$ converges in distribution and in all moments to a mixed Poisson variable whose mean is an explicit function of a gamma variable with shape r. These limits depend on r and, after standardization, converge to the interior law as $r\to\infty$. Together with the Erlang weighted tree of Moharrami, Subramanian, Liu and Sundaresan for fixed k, this identifies every subsequential limit of the standardized degree; no single law serves for all sequences $k(n)$. We also show that $\mathbb{E}(k-D)$ takes the same value at the budgets k and $n-1-k$ for every n.

1. Introduction

Let $n\geq 3$ and $1\leq k\leq n-1$. Put independent scores $U_{ij}=U_{ji}$, uniform on $[0,1]$, on the edges of the complete graph K_n. Each vertex prefers the k edges at it with the smallest scores, and the graph $LK(n,k)$ keeps an edge if and only if both of its endpoints prefer it. The model is due to La and Kabkab [4], who studied its connectivity and diameter. Dabirian and Subramanian [2] let each vertex prefer its k edges of largest score; the map $U\mapsto 1-U$ turns one description into the other, so the two give the same law. Moharrami, Subramanian, Liu and Sundaresan [6] study a finite model with independent exponential edge costs in which every vertex is willing to use its d_i cheapest edges; when every d_i equals k this is again $LK(n,k)$, since only the order of the scores matters.

Fix a vertex 0, the root, and let $D=D_{n,k}$ be its degree. The root keeps only edges it prefers, so $D\leq k$. Dabirian and Subramanian show that

$$\mathbb{E}D=k-c_k\bigl(1+O(k^2/n)\bigr),\qquad c_k=\frac{(2k-1)\binom{2k-2}{k-1}}{2^{2k-1}},$$

when $k=o(\sqrt{n})$ [2, Theorem 7], and that $\mathbb{E}D=k-\sqrt{k/\pi}+1/(8\sqrt{\pi k})+o(k^{-1/2})$ when $k\to\infty$ with $k=o(n^{1/3})$ [2, Theorem 8]. Among their open problems [2, Section 1.1] they ask for the limiting distribution of $(D-\mathbb{E}D)/\sqrt{\operatorname{Var} D}$, and they note that even $\operatorname{Var} D$ is hard to compute, because the decisions of different vertices are dependent. For fixed k the answer is contained in [6]: the graph converges locally to the Erlang weighted tree [6, Theorem 2.1], and the law of the root degree of that tree is given in [6, Theorem 5.1]. We treat the remaining regimes.

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Write $H = k - D$; it counts the edges preferred by the root and not by their other endpoint. Put

$$s_n^2 = \frac{k(n-k)}{n}, \qquad g(z) = \phi(z) + z\Phi(z) = \mathbb{E}(z - Z')_+,$$
$$v = \operatorname{Var} g(Z) = \frac{1}{3} + \frac{\sqrt{3}-2}{2\pi} = 0.290688\ldots,$$

where ϕ and Φ are the standard normal density and distribution function and Z, Z' are independent standard normal variables. The results are as follows.

(1) *Interior.* If $k \to \infty$ and $n - k \to \infty$, then $H/s_n \Rightarrow g(Z)$ with convergence of the first two moments (Theorem 2.1). Hence $\mathbb{E}D = k - s_n/\sqrt{\pi} + o(s_n)$, $\operatorname{Var} D = v s_n^2 + o(s_n^2)$, and $(D - \mathbb{E}D)/\sqrt{\operatorname{Var} D}$ converges in distribution to $L_\infty = (\pi^{-1/2} - g(Z))/\sqrt{v}$. This law has an explicit density (Section 3) and takes values in $(-\infty, 1.0464\ldots)$, so it is not normal. In the regime of [2, Theorem 8] the formula for $\mathbb{E}D$ agrees with theirs to first order (Remark 2.6).
(2) *Fixed complement.* If $r = n - 1 - k \geq 1$ is fixed, then H converges in distribution and in all moments to a variable H_r which, given a gamma variable A_r with shape r and scale 1, is Poisson with mean $\Lambda_r(A_r) = e^{-A_r} \sum_{\ell<r} (r-\ell) A_r^\ell/\ell!$ (Theorem 4.2). Its mean is $r\binom{2r}{r}4^{-r}$ (Proposition 4.3). The standardized limits for $r = 1$ and $r = 2$ are different, and as $r \to \infty$ they converge to L_∞ (Theorem 5.1).
(3) *All sequences.* Along any sequence $k(n)$, every subsequential limit of the standardized degree is one of these laws or a fixed-k law from [6], and no single law serves for all sequences (Theorem 6.2).
(4) *Duality.* For every n and $1 \leq k \leq n-2$, $\mathbb{E}H$ is the same at the budgets k and $n-1-k$ (Proposition 7.1). As $n \to \infty$ this identifies the constant c_k of [2, Theorem 7] with the mean of H_k (Corollary 7.3).

The dependence that makes $\operatorname{Var} D$ hard is weak once the root's own cutoff is fixed. Let X be the $(k+1)$st smallest score at the root. Given X, two neighbours of the root interact only through the edge between them, and Lemma 2.2 bounds the resulting covariance by $1/(4(n-2)^2X^2)$, which is $O(k^{-2})$ when X is of order k/n. Summed over the k^2 pairs this is $O(1)$, so H stays within $O(s_n^{1/2})$ of its conditional mean $kq_n(X)$, a function of X alone, and the limit $g(Z)$ is that function seen at the scale of the fluctuations of X. Along the way Lemma 2.3 gives $\operatorname{Var} D$ exactly for every n and k. At a fixed complement the relevant scale is $n(1-U)$ at the top of the root's scores; there the candidates form a Poisson process in the limit, and the finitely many edges among them each matter with probability $O(1/n)$. Section 8 describes the computer checks that accompany the paper.

2. The interior regime

Theorem 2.1. *Suppose that $k = k(n) \to \infty$ and $n - k \to \infty$. Then $H/s_n \Rightarrow g(Z)$, and*

$$\frac{\mathbb{E}H}{s_n} \to \mathbb{E}g(Z) = \frac{1}{\sqrt{\pi}}, \qquad \frac{\mathbb{E}H^2}{s_n^2} \to \mathbb{E}g(Z)^2 = \frac{1}{3} + \frac{\sqrt{3}}{2\pi}.$$

Consequently $\mathbb{E}D = k - s_n/\sqrt{\pi} + o(s_n)$, $\operatorname{Var} D = v s_n^2 + o(s_n^2)$, and

$$\frac{D - \mathbb{E}D}{\sqrt{\operatorname{Var} D}} \Longrightarrow L_\infty := \frac{\pi^{-1/2} - g(Z)}{\sqrt{v}}.$$

2.1. **A representation of H.** Let $1 \leq k \leq n-2$; for $k = n-1$ every edge is kept and $H = 0$. Let X be the $(k+1)$st smallest of the $n-1$ scores at the root and let S be the set of the k vertices i with $U_{0i} < X$. Then $X \sim \operatorname{Beta}(k+1, n-k-1)$. Given X and S, the scores U_{0i}, $i \in S$,

are independent and uniform on $(0, X)$, and they are independent of the scores of the edges not at the root, which are still independent and uniform on $[0, 1]$.

For $i \in S$ let T_i be the kth smallest of the $n-2$ scores U_{ij}, $j \notin \{0, i\}$. Then $T_i \sim \mathrm{Beta}(k, n-k-1)$, and i prefers the edge $\{0, i\}$ if and only if $U_{0i} < T_i$. With $R_i = \mathbf{1}\{U_{0i} > T_i\}$ we have $H = \sum_{i \in S} R_i$. Let T be a $\mathrm{Beta}(k, n-k-1)$ variable independent of X, let F_T and f_T be its distribution function and density, and put

$$q_n(x) = \frac{1}{x}\,\mathbb{E}(x - T)_+ = \frac{1}{x}\int_0^x F_T(u)\,du, \qquad 0 < x \le 1. \tag{2.1}$$

Given X, each U_{0i} is uniform on $(0, X)$ and independent of T_i, so $\mathbb{P}(R_i = 1 \mid X) = q_n(X)$ and

$$\mathbb{E}[H \mid X] = k q_n(X). \tag{2.2}$$

None of these conditional laws depends on S.

2.2. **Two neighbours of the root.**

Lemma 2.2. *Let $1 \le k \le n-2$, let $i, j \in S$ be distinct, and write $W = U_{ij}$. Given X and W, the indicators R_i and R_j are independent, and $\mathbb{P}(R_i = 1 \mid X, W) = p_X(W)$, where*

$$p_x(w) = b_x + \frac{(F_T(x) - F_T(w))_+}{(n-2)x}, \qquad b_x = \frac{1}{x}\int_0^x \mathbb{P}(\mathrm{Bin}(n-3, u) \ge k)\,du.$$

Moreover $\mathbb{E}[p_X(W) \mid X] = q_n(X)$ and

$$0 \le \mathrm{Cov}(R_i, R_j \mid X) = \mathrm{Var}(p_X(W) \mid X) \le \frac{F_T(X)^2}{4(n-2)^2X^2} \le \frac{1}{4(n-2)^2X^2}.$$

Proof. Given X, the vertex i decides about $\{0, i\}$ using U_{0i}, W and the $n-3$ scores U_{il} with $l \notin \{0, i, j\}$, and j uses U_{0j}, W and the scores U_{jl} with $l \notin \{0, i, j\}$. Apart from W these sets of scores are disjoint, and U_{0i}, U_{0j} are independent and uniform on $(0, X)$ given X and S. So R_i and R_j are independent given (X, W), by symmetry both have conditional probability $p_X(W)$, and since W is independent of X, $\mathrm{Cov}(R_i, R_j \mid X) = \mathrm{Var}(p_X(W) \mid X)$.

Fix $U_{0i} = u < X$ and $W = w$, and let N_u be the number of $l \notin \{0, i, j\}$ with $U_{il} < u$, so that $N_u \sim \mathrm{Bin}(n-3, u)$. The vertex i rejects $\{0, i\}$ if and only if at least k of its other $n-2$ scores lie below u, that is, $N_u + \mathbf{1}\{w < u\} \ge k$. Hence

$$\mathbb{P}(R_i = 1 \mid U_{0i} = u,\, W = w) = \mathbb{P}(N_u \ge k) + \mathbf{1}\{w < u\}\,\mathbb{P}(N_u = k-1).$$

Since $\binom{n-3}{k-1}B(k, n-k-1) = 1/(n-2)$, we have $\mathbb{P}(N_u = k-1) = \binom{n-3}{k-1}u^{k-1}(1-u)^{n-k-2} = f_T(u)/(n-2)$. Averaging over u uniform on $(0, X)$ gives the formula for $p_X(w)$, and averaging $p_X(W)$ over W gives $\mathbb{P}(R_i = 1 \mid X) = q_n(X)$. Finally $p_X(W) - b_X$ takes values in $[0, F_T(X)/((n-2)X)]$, and a random variable with values in an interval of length ℓ has variance at most $\ell^2/4$ (Popoviciu's inequality). □

Lemma 2.3. *For $1 \le k \le n-2$,*

$$\mathrm{Var}\,H = \mathbb{E}\big[k q_n(X)(1 - q_n(X))\big] + k(k-1)\,\mathbb{E}\big[\mathrm{Var}(p_X(W) \mid X)\big] + k^2\,\mathrm{Var}\,q_n(X).$$

In particular

$$\mathrm{Var}(H \mid X) \le k q_n(X) + \frac{k^2}{4(n-2)^2X^2}. \tag{2.3}$$

Proof. Given X, H is a sum of k indicators with mean $q_n(X)$ whose pairwise covariances equal $\mathrm{Var}(p_X(W) \mid X)$ by Lemma 2.2. So $\mathrm{Var}(H \mid X) = k q_n(X)(1 - q_n(X)) + k(k-1)\,\mathrm{Var}(p_X(W) \mid X)$, and the identity follows from $\mathrm{Var}\,H = \mathbb{E}\,\mathrm{Var}(H \mid X) + \mathrm{Var}\,\mathbb{E}[H \mid X]$ and (2.2). The bound follows from $q_n(1 - q_n) \le q_n$ and Lemma 2.2. □

All three terms are explicit integrals against the beta laws of X and T, so Lemma 2.3 gives $\operatorname{Var} D = \operatorname{Var} H$ in closed form for every n and k; Section 8 uses it.

2.3. **The variance given X is small.**

Lemma 2.4. *Let $n \geq 4$, $1 \leq k \leq n-2$, $G_n = \{X \geq k/(2n)\}$ and $c = \log 2 - 1/2$. Then $\mathbb{P}(G_n^c) \leq e^{-ck}$ and*

$$\mathbb{E}\big[(H - kq_n(X))^2\big] = \mathbb{E}\operatorname{Var}(H \mid X) \leq 6 + 4s_n + k^2 e^{-ck}.$$

If $k \to \infty$ and $n - k \to \infty$, then $\mathbb{E}[(H - kq_n(X))^2] = o(s_n^2)$.

Proof. The event $X < k/(2n)$ says that at least $k+1$ of the $n-1$ scores at the root lie below $k/(2n)$, so $\mathbb{P}(G_n^c) = \mathbb{P}(\mathrm{Bin}(n-1, k/(2n)) \geq k+1)$. For a binomial variable N with mean μ and $t > 0$, $\mathbb{P}(N \geq m) \leq e^{-tm}\mathbb{E}e^{tN} \leq \exp(-tm + \mu(e^t - 1))$; with $t = \log 2$, $m = k+1$ and $\mu \leq k/2$ this is at most $2^{-k-1}e^{k/2} \leq e^{-ck}$.

On G_n the second term in (2.3) is at most $n^2/(n-2)^2 \leq 4$. For the first, $(x-t)_+ \leq |x-t|$ gives $kq_n(X) \leq (k/X)\mathbb{E}[|X-T| \mid X] \leq 2n\,\mathbb{E}[|X-T| \mid X]$ on G_n, and therefore

$$\mathbb{E}[kq_n(X); G_n] \leq 2n\,\mathbb{E}|X-T| \leq 2n\big(\operatorname{Var} X + \operatorname{Var} T + (\mathbb{E}X - \mathbb{E}T)^2\big)^{1/2}.$$

The beta moments give $\operatorname{Var} X = (k+1)(n-k-1)/(n^2(n+1)) \leq 2s_n^2/n^2$, $\operatorname{Var} T = k(n-k-1)/((n-1)^2 n) \leq 2s_n^2/n^2$ for $n \geq 4$, and $0 \leq \mathbb{E}X - \mathbb{E}T = (n-1-k)/(n(n-1)) \leq 1/n$. Hence $\mathbb{E}[kq_n(X); G_n] \leq 2n(4s_n^2/n^2 + 1/n^2)^{1/2} \leq 4s_n + 2$. On G_n^c we use $\operatorname{Var}(H \mid X) \leq k^2$, which holds because $0 \leq H \leq k$. Adding up gives the bound.

For the last claim note that $s_n^2 \geq \min(k, n-k)/2 \to \infty$, so $(6 + 4s_n)/s_n^2 \to 0$. If $k \leq n/2$ then $s_n^2 \geq k/2$ and $k^2 e^{-ck}/s_n^2 \leq 2ke^{-ck} \to 0$. If $k > n/2$ then $k^2 e^{-ck} \leq n^2 e^{-cn/2} \to 0$ while $s_n^2 \geq 1/2$. □

2.4. **The conditional mean.** Put $p_n = k/n$, $\sigma_n = s_n/n$,

$$U_n = \frac{X - p_n}{\sigma_n}, \qquad V_n = \frac{T - p_n}{\sigma_n}, \qquad g_n(u) = \mathbb{E}(u - V_n)_+.$$

Since $(X-T)_+ = \sigma_n(U_n - V_n)_+$ and $k\sigma_n/s_n = k/n$,

$$\frac{kq_n(X)}{s_n} = \frac{k}{nX}\, g_n(U_n). \tag{2.4}$$

Lemma 2.5. *If $k \to \infty$ and $n - k \to \infty$, then $kq_n(X)/s_n \Rightarrow g(Z)$, and the squares $(kq_n(X)/s_n)^2$ are uniformly integrable. In particular $\mathbb{E}[kq_n(X)]/s_n \to \mathbb{E}g(Z)$ and $\mathbb{E}[(kq_n(X))^2]/s_n^2 \to \mathbb{E}g(Z)^2$.*

Proof. Normal limits. Fix $u \in \mathbb{R}$ and put $\pi_n = p_n + \sigma_n u$. Since $\sigma_n/p_n = s_n/k \leq k^{-1/2}$ and $\sigma_n/(1-p_n) = s_n/(n-k) \leq (n-k)^{-1/2}$, for large n we have $0 < \pi_n < 1$ and $(n-1)\pi_n(1-\pi_n) = s_n^2(1 + o(1)) \to \infty$. Now $\mathbb{P}(U_n \leq u) = \mathbb{P}(X \leq \pi_n) = \mathbb{P}(\mathrm{Bin}(n-1, \pi_n) \geq k+1)$, and this binomial variable has mean $k + s_n u + O(1 + |u|)$. By the central limit theorem for binomial variables whose variance tends to infinity, $\mathbb{P}(U_n \leq u) \to \mathbb{P}(Z \geq -u) = \Phi(u)$. In the same way $\mathbb{P}(V_n \leq u) = \mathbb{P}(\mathrm{Bin}(n-2, \pi_n) \geq k) \to \Phi(u)$. So $U_n \Rightarrow Z$ and $V_n \Rightarrow Z'$.

Fourth moments. For a $\mathrm{Beta}(a, b)$ variable the fourth central moment is $3 + \kappa$ times the squared variance, where

$$\kappa = \frac{6\big[(a-b)^2(a+b+1) - ab(a+b+2)\big]}{ab(a+b+2)(a+b+3)}, \qquad |\kappa| \leq \frac{6}{a} + \frac{6}{b} + \frac{6}{a+b}.$$

For $(a, b) = (k+1, n-k-1)$ and $(k, n-k-1)$ this tends to 0. Moreover $\operatorname{Var} X/\sigma_n^2 \to 1$, $\operatorname{Var} T/\sigma_n^2 \to 1$, $(\mathbb{E}X - p_n)/\sigma_n = 1/s_n \to 0$ and $(\mathbb{E}T - p_n)/\sigma_n = k/((n-1)s_n) \to 0$. Hence $\mathbb{E}U_n^4 \to 3$ and $\mathbb{E}V_n^4 \to 3$.

Uniform convergence of g_n. By the Skorokhod representation we may take $V_n \to Z'$ almost surely. The variables $|V_n - Z'|$ are uniformly integrable because $\sup_n \mathbb{E}V_n^2 < \infty$, and $x \mapsto (u-x)_+$ is 1-Lipschitz, so $\sup_u |g_n(u) - g(u)| \le \mathbb{E}|V_n - Z'| \to 0$.

Convergence in law. Since $X/p_n = 1 + (\sigma_n/p_n)U_n$ and U_n is tight, $k/(nX) = p_n/X \to 1$ in probability. As g is continuous, $g(U_n) \Rightarrow g(Z)$, and with the uniform convergence of g_n, (2.4) and Slutsky's lemma, $kq_n(X)/s_n \Rightarrow g(Z)$.

Uniform integrability. On G_n we have $k/(nX) \le 2$, and $g_n(u) \le |u| + \mathbb{E}|V_n| \le |u| + 2$ for large n, so $(kq_n(X)/s_n)^2 \le 4(|U_n|+2)^2$; these bounds are uniformly integrable because $\sup_n \mathbb{E}U_n^4 < \infty$. On G_n^c we have $kq_n(X) \le k$, so $\mathbb{E}[(kq_n(X)/s_n)^2; G_n^c] \le k^2 e^{-ck}/s_n^2 \to 0$, as in the proof of Lemma 2.4. □

Proof of Theorem 2.1. By Lemma 2.4, $(H - kq_n(X))/s_n \to 0$ in L^2. With Lemma 2.5 this gives $H/s_n \Rightarrow g(Z)$, and since the L^2 norms of H/s_n and $kq_n(X)/s_n$ differ by $o(1)$, also $\mathbb{E}H/s_n \to \mathbb{E}g(Z)$ and $\mathbb{E}H^2/s_n^2 \to \mathbb{E}g(Z)^2$.

For the constants, $Z - Z'$ is normal with variance 2, so $\mathbb{E}g(Z) = \mathbb{E}(Z - Z')_+ = 1/\sqrt{\pi}$. With Z_1, Z_2 independent copies of Z', $g(Z)^2 = \mathbb{E}[(Z - Z_1)_+(Z - Z_2)_+ \mid Z]$, and $(Z - Z_1, Z - Z_2)/\sqrt{2}$ is a standard normal pair with correlation $1/2$. For a standard normal pair (ξ, η) with correlation ρ,

$$\mathbb{E}[\xi_+\eta_+] = \frac{\sqrt{1-\rho^2} + \rho(\pi/2 + \arcsin\rho)}{2\pi},$$

which is $1/6 + \sqrt{3}/(4\pi)$ at $\rho = 1/2$. Hence $\mathbb{E}g(Z)^2 = 1/3 + \sqrt{3}/(2\pi)$ and $\operatorname{Var} g(Z) = \mathbb{E}g(Z)^2 - 1/\pi = v$. Since $D = k - H$, we get $\mathbb{E}D = k - s_n/\sqrt{\pi} + o(s_n)$ and $\operatorname{Var} D = \operatorname{Var} H = vs_n^2 + o(s_n^2)$, and

$$\frac{D - \mathbb{E}D}{\sqrt{\operatorname{Var} D}} = \frac{\mathbb{E}H/s_n - H/s_n}{\sqrt{\operatorname{Var} H/s_n^2}} \Longrightarrow \frac{\pi^{-1/2} - g(Z)}{\sqrt{v}}.$$ □

Remark 2.6. If $k \to \infty$ and $k = o(n^{1/3})$, the regime of [2, Theorem 8], then $0 \le \sqrt{k} - s_n \le \sqrt{k} \cdot k/n = k^{3/2}/n \to 0$, so Theorem 2.1 gives $\mathbb{E}D = k - \sqrt{k/\pi} + o(\sqrt{k})$, the first two terms of [2, Theorem 8], and the standardized degree converges to L_∞ in that regime as well.

3. The interior limit law

Since $g' = \Phi > 0$, $g(z) \to 0$ as $z \to -\infty$ and $g(z) - z \to 0$ as $z \to \infty$, the function g increases strictly from 0 to ∞. Let $m = \pi^{-1/2}$. The variable $L_\infty = (m - g(Z))/\sqrt{v}$ has mean 0 and variance 1 and takes values in $(-\infty, m/\sqrt{v})$, where $m/\sqrt{v} = 1.04643\ldots$. For $x < m/\sqrt{v}$ let $y_x = g^{-1}(m - \sqrt{v}\,x)$. Then

$$\mathbb{P}(L_\infty \le x) = \mathbb{P}(Z \ge y_x) = 1 - \Phi(y_x), \tag{3.1}$$

and differentiating with $g' = \Phi$ gives the density

$$f_\infty(x) = \frac{\sqrt{v}\,\phi(y_x)}{\Phi(y_x)}, \qquad x < \frac{m}{\sqrt{v}}. \tag{3.2}$$

As $x \to -\infty$ we have $y_x = m - \sqrt{v}\,x + o(1)$, so the lower tail is Gaussian. As x increases to $m/\sqrt{v}$, $y_x \to -\infty$ and $f_\infty(x) \sim \sqrt{v}\,|y_x| \to \infty$. The law is bounded above, so it is not normal.

4. The fixed complement

Fix $r \ge 1$ and let $k = n - 1 - r$, so that every vertex rejects exactly its r edges of largest score. For $i \ne 0$ put $Y_i = n(1 - U_{0i})$; these are independent and uniform on $(0, n)$. Let $A_{n,r}$ be the rth smallest of $Y_1, \ldots, Y_{n-1}$. The root prefers $\{0, i\}$ if and only if $Y_i > A_{n,r}$. Let J_i be the indicator

that i rejects $\{0,i\}$, which happens if and only if fewer than r of the other $n-2$ scores at i exceed U_{0i}. Then $H=\sum_{i\neq 0}\mathbf{1}\{Y_i>A_{n,r}\}J_i$ and

$$\mathbb{P}(J_i=1\mid Y_i=y)=\mathbb{P}(\mathrm{Bin}(n-2,y/n)\le r-1)\to p_r(y):=e^{-y}\sum_{\ell<r}\frac{y^\ell}{\ell!}=\mathbb{P}(G>y),$$

where G has the gamma law with shape r and scale 1, written $\Gamma(r,1)$.

In the limit the points Y_i form a Poisson process of rate 1 on $(0,\infty)$, the root rejects the lowest r of them, and the others are marked independently, a point at y with probability $p_r(y)$. Given that the rth point is at a, the marked points above a then form a Poisson process with intensity p_r on (a,∞), so their number is Poisson with mean

$$\Lambda_r(a)=\int_a^\infty p_r(y)\,dy=\mathbb{E}(G-a)_+=e^{-a}\sum_{\ell<r}(r-\ell)\frac{a^\ell}{\ell!}. \tag{4.1}$$

The last equality is $\int_a^\infty e^{-y}y^i/i!\,dy=e^{-a}\sum_{\ell\le i}a^\ell/\ell!$, summed over $i<r$. We prove this description by factorial moments; the one dependence to control is the shared edge between two candidates.

Lemma 4.1. *Let $j\ge 1$, let $i_1,\dots,i_j$ be distinct vertices other than 0, and let $y\in(0,n)^j$. Let $Q_n(y)$ be the probability that every i_a rejects $\{0,i_a\}$ given $Y_{i_a}=y_a$ for all a, and let $\beta_n(y)=\mathbb{P}(\mathrm{Bin}(n-1-j,y/n)\le r-1)$. Then*

$$0\le\prod_{a=1}^j\beta_n(y_a)-Q_n(y)\le\binom{j}{2}\frac{\max_a y_a}{n}.$$

Proof. Split the $n-2$ edges at i_a other than $\{0,i_a\}$ into the $j-1$ edges to the other i_b and the $n-1-j$ edges to the remaining vertices, and let C_a be the number of the latter whose score exceeds $1-y_a/n$. The C_a are independent with $C_a\sim\mathrm{Bin}(n-1-j,y_a/n)$, and i_a rejects $\{0,i_a\}$ if and only if C_a plus the number of edges $\{i_a,i_b\}$ with score above $1-y_a/n$ is at most $r-1$. So rejection by every i_a forces $C_a\le r-1$ for every a, which gives the first inequality. If every edge $\{i_a,i_b\}$ has score at most $1-\max(y_a,y_b)/n$, the two events coincide, and each of the $\binom{j}{2}$ edges fails this with probability at most $\max_a y_a/n$. □

Theorem 4.2. *Fix $r\ge 1$ and let $k=n-1-r$. Let $A_r\sim\Gamma(r,1)$ and, given A_r, let H_r be Poisson with mean $\Lambda_r(A_r)$. Then H converges to H_r in distribution and in all moments. For every $j\ge 1$,*

$$\mathbb{E}[(H)_j]\to\mathbb{E}[(H_r)_j]=\mathbb{E}[\Lambda_r(A_r)^j],$$

where $(h)_j=h(h-1)\cdots(h-j+1)$. The generating function of H_r is

$$\mathbb{E}s^{H_r}=\frac{1}{(r-1)!}\int_0^\infty a^{r-1}e^{-a}\exp\bigl(-(1-s)\Lambda_r(a)\bigr)\,da,\qquad 0\le s\le 1.$$

Proof. For an ordered j-tuple of distinct vertices $i_1,\dots,i_j$, the root prefers all of them if and only if at least r of the other $n-1-j$ values Y_l lie below $\min_a Y_{i_a}$. Given $Y_{i_a}=y_a$ for all a, this event depends only on the other scores at the root, and the rejections depend only on scores not at the root, so the two are conditionally independent. Summing over ordered j-tuples,

$$\mathbb{E}[(H)_j]=\frac{(n-1)_j}{n^j}\int_{(0,n)^j}P_n(y)Q_n(y)\,dy,\qquad P_n(y)=\mathbb{P}\Bigl(\mathrm{Bin}\bigl(n-1-j,\min_a y_a/n\bigr)\ge r\Bigr). \tag{4.2}$$

Fix $y\in(0,\infty)^j$. As $n\to\infty$, $P_n(y)\to\mathbb{P}(\mathrm{Poi}(\min_a y_a)\ge r)=\mathbb{P}(A_r<\min_a y_a)$, $\beta_n(y_a)\to p_r(y_a)$, hence $Q_n(y)\to\prod_a p_r(y_a)$ by Lemma 4.1, and $(n-1)_j/n^j\to 1$. For a dominating function, let

$n \ge 2(r+j)$ and $i \le r-1$; then

$$\mathbb{P}(\mathrm{Bin}(n-1-j, y/n) = i) \le \frac{y^i}{i!}\Big(1-\frac{y}{n}\Big)^{n-1-j-i} \le \frac{y^i}{i!}e^{-y/2},$$

so $P_n Q_n \le \prod_a \beta_n(y_a) \le \prod_a \psi_r(y_a)$ with $\psi_r(y) = e^{-y/2}\sum_{i<r} y^i/i!$, which is integrable on $(0,\infty)$. By dominated convergence and Fubini's theorem,

$$\mathbb{E}[(H)_j] \to \int_{(0,\infty)^j} \mathbb{P}\Big(A_r < \min_a y_a\Big)\prod_a p_r(y_a)\,dy = \mathbb{E}\Big[\prod_{a=1}^{j}\int_{A_r}^{\infty} p_r(y_a)\,dy_a\Big] = \mathbb{E}[\Lambda_r(A_r)^j].$$

A Poisson variable with mean λ has jth factorial moment λ^j, so the limit is $\mathbb{E}[(H_r)_j]$.

Since Λ_r decreases, $\Lambda_r \le \Lambda_r(0) = \mathbb{E}G = r$. Hence $\mathbb{E}[(H_r)_j] \le r^j$ and $\mathbb{E}e^{tH_r} \le \exp(r(e^t-1))$ for every $t>0$, so the law of H_r is determined by its moments [1, Theorem 30.1]. Every power H^m is a fixed linear combination of $(H)_1,\dots,(H)_m$, so all moments of H converge to those of H_r, and the method of moments [1, Theorem 30.2] gives $H \Rightarrow H_r$. The formula for $\mathbb{E}s^{H_r}$ is the Poisson generating function averaged over A_r. □

4.1. **Moments.** Let $\mu_r = \mathbb{E}H_r$, $\nu_r = \operatorname{Var} H_r$ and $M_{2,r} = \mathbb{E}\Lambda_r(A_r)^2$. Since H_r is Poisson given A_r,

$$\mu_r = \mathbb{E}\Lambda_r(A_r), \qquad \nu_r = \mu_r + \operatorname{Var}\Lambda_r(A_r) = \mu_r + M_{2,r} - \mu_r^2 > 0.$$

By Theorem 4.2, $\mathbb{E}D = k - \mu_r + o(1)$, $\operatorname{Var} D = \nu_r + o(1)$ and

$$\frac{D-\mathbb{E}D}{\sqrt{\operatorname{Var} D}} \Longrightarrow L_r := \frac{\mu_r - H_r}{\sqrt{\nu_r}}.$$

For $A \sim \Gamma(r,1)$, $c>0$ and an integer $N \ge 0$, $\mathbb{E}[e^{-cA}A^N] = (r-1+N)!/((r-1)!(1+c)^{r+N})$. Expanding (4.1) therefore gives the finite sums

$$\mu_r = \sum_{\ell<r} \frac{(r-\ell)\,(r-1+\ell)!}{\ell!\,(r-1)!\,2^{r+\ell}}, \tag{4.3}$$

$$M_{2,r} = \frac{1}{(r-1)!}\sum_{\ell,m<r} \frac{(r-\ell)(r-m)\,(r-1+\ell+m)!}{\ell!\,m!\,3^{r+\ell+m}}. \tag{4.4}$$

Proposition 4.3. *For every $r \ge 1$,*

$$\mu_r = \frac{r\binom{2r}{r}}{4^r} = \frac{\Gamma(r+1/2)}{\sqrt{\pi}\,\Gamma(r)}.$$

Proof. Let A, G be independent $\Gamma(r,1)$ variables. By (4.1), $\mu_r = \mathbb{E}(G-A)_+ = \frac12\mathbb{E}|G-A|$. The variables $S = A+G \sim \Gamma(2r,1)$ and $B = A/S \sim \mathrm{Beta}(r,r)$ are independent, and $G-A = S(1-2B)$, so $\mu_r = \frac12\mathbb{E}S\cdot\mathbb{E}|1-2B| = r\,\mathbb{E}|1-2B|$. The variable $1-2B$ has density $(1-w^2)^{r-1}/(2^{2r-1}B(r,r))$ on $(-1,1)$, whence $\mathbb{E}|1-2B| = 1/(r\,2^{2r-1}B(r,r))$ and

$$\mu_r = \frac{1}{2^{2r-1}B(r,r)} = \frac{2\,\Gamma(2r)}{4^r\,\Gamma(r)^2} = \frac{r\binom{2r}{r}}{4^r}.$$

The duplication formula $\Gamma(2r) = 2^{2r-1}\Gamma(r)\Gamma(r+1/2)/\sqrt{\pi}$ gives the last form. □

For $r=1$ and $r=2$, (4.3) and (4.4) give

$$\mu_1 = \tfrac12, \quad M_{2,1} = \tfrac13, \quad \nu_1 = \tfrac{7}{12}; \qquad \mu_2 = \tfrac34, \quad M_{2,2} = \tfrac{22}{27}, \quad \nu_2 = \tfrac{433}{432}. \tag{4.5}$$

Since $\mathbb{P}(H_r = 0) > 0$, the largest value of L_r is $\mu_r/\sqrt{\nu_r}$, whose square is $3/7$ for $r=1$ and $243/433$ for $r=2$. So L_1 and L_2 are different laws, with largest values $0.6547\ldots$ and $0.7491\ldots$.

5. From the fixed complement to the interior

Theorem 5.1. *As $r \to \infty$, $H_r/\sqrt{r} \Rightarrow g(Z)$ with convergence of the first two moments. Consequently $\mu_r/\sqrt{r} \to 1/\sqrt{\pi}$, $\nu_r/r \to v$ and $L_r \Rightarrow L_\infty$.*

Proof. Let A_r, G_r be independent $\Gamma(r,1)$ variables and put $U_r = (A_r - r)/\sqrt{r}$, $V_r = (G_r - r)/\sqrt{r}$ and $h_r(u) = \mathbb{E}(V_r - u)_+$. By (4.1), $\Lambda_r(A_r)/\sqrt{r} = h_r(U_r)$. By the central limit theorem $U_r, V_r \Rightarrow N(0,1)$, and $\mathbb{E}V_r^2 = 1$, so as in the proof of Lemma 2.5 h_r converges uniformly to $h(u) = \mathbb{E}(Z' - u)_+ = g(-u)$. Hence $h_r(U_r) \Rightarrow g(-Z)$, which has the law of $g(Z)$. Moreover $0 \le h_r(u) \le |u| + 1$ and $\mathbb{E}U_r^4 = 3 + 6/r$, so the squares $h_r(U_r)^2$ are uniformly integrable, and $\mathbb{E}\Lambda_r(A_r)/\sqrt{r} \to 1/\sqrt{\pi}$, $\mathbb{E}\Lambda_r(A_r)^2/r \to \mathbb{E}g(Z)^2$. In particular $\mu_r = O(\sqrt{r})$. The Poisson noise is negligible:

$$\frac{1}{r}\mathbb{E}\big[(H_r - \Lambda_r(A_r))^2\big] = \frac{\mathbb{E}\Lambda_r(A_r)}{r} = \frac{\mu_r}{r} \to 0.$$

So $H_r/\sqrt{r} \Rightarrow g(Z)$ with convergence of the first two moments, which gives $\mu_r/\sqrt{r} \to 1/\sqrt{\pi}$ and $\nu_r/r \to v$, and $L_r = (\mu_r/\sqrt{r} - H_r/\sqrt{r})/\sqrt{\nu_r/r} \Rightarrow L_\infty$. □

6. Fixed k and all sequences $k(n)$

For fixed k the root degree converges to the root degree of the Erlang weighted tree. The local limit is [6, Theorem 2.1] and the law of the root degree is [6, Theorem 5.1]; the following short proof, in the notation of Section 2, gives the form we use.

Proposition 6.1. *Fix $k \ge 1$. Let $X_\infty \sim \Gamma(k+1,1)$ and $T_\infty \sim \Gamma(k,1)$, put $q_\infty(x) = \mathbb{E}(x - T_\infty)_+/x$, and let D_k^∞ have, given $X_\infty = x$, the law $\mathrm{Bin}(k, 1 - q_\infty(x))$. Then $D_{n,k} \Rightarrow D_k^\infty$ as $n \to \infty$, with convergence of all moments, and $\operatorname{Var} D_k^\infty > 0$.*

Proof. Let $n \ge k + 2$. For $i \in S$, the scores at i other than U_{0i} are those of the $k - 1$ edges to the rest of S and those of the $n - 1 - k$ edges from i to vertices outside $S \cup \{0\}$; for different i the second groups are disjoint. Let E_n be the event that the $\binom{k}{2}$ edges inside S all have scores larger than X. On E_n, i rejects $\{0, i\}$ if and only if at least k of its scores to vertices outside $S \cup \{0\}$ are below U_{0i}. Let $\tilde{R}_i$ be the indicator of this last event and $\tilde{H} = \sum_{i \in S} \tilde{R}_i$. Given X, the $\tilde{R}_i$ are independent with

$$\mathbb{P}(\tilde{R}_i = 1 \mid X) = \tilde{b}_n(X), \qquad \tilde{b}_n(x) = \frac{1}{x}\int_0^x \mathbb{P}(\mathrm{Bin}(n - 1 - k, u) \ge k)\, du,$$

so $\tilde{H}$ is $\mathrm{Bin}(k, \tilde{b}_n(X))$ given X. The scores inside S are independent of X, so $\mathbb{P}(H \ne \tilde{H}) \le \mathbb{P}(E_n^c) \le \binom{k}{2}\mathbb{E}X = \binom{k}{2}(k+1)/n$.

Substituting $u = y/n$, $\tilde{b}_n(X) = (nX)^{-1}\int_0^{nX} \mathbb{P}(\mathrm{Bin}(n-1-k, y/n) \ge k)\, dy$. Le Cam's inequality and the bound $|a - b|$ for the total variation distance between Poisson laws with means a and b give, for $0 < y < n$,

$$\big|\mathbb{P}(\mathrm{Bin}(n - 1 - k, y/n) \ge k) - \mathbb{P}(\mathrm{Poi}(y) \ge k)\big| \le \frac{y^2}{n} + \frac{(k+1)y}{n},$$

and $\mathbb{P}(\mathrm{Poi}(y) \ge k) = \mathbb{P}(T_\infty \le y)$. Hence $|\tilde{b}_n(X) - q_\infty(nX)| \le (nX)^2/(3n) + (k+1)nX/(2n)$, which tends to 0 in probability because $nX \Rightarrow X_\infty$. As q_∞ is continuous, $\tilde{b}_n(X) \Rightarrow q_\infty(X_\infty)$, and applying the bounded continuous functions $p \mapsto \binom{k}{h}p^h(1-p)^{k-h}$ shows that $\tilde{H}$ converges in law to $k - D_k^\infty$. So $D_{n,k} = k - H \Rightarrow D_k^\infty$, and the moments converge because $0 \le D_{n,k} \le k$. Finally $\operatorname{Var} D_k^\infty \ge \mathbb{E}[k q_\infty(X_\infty)(1 - q_\infty(X_\infty))] > 0$, since $0 < q_\infty(x) < 1$ for $x > 0$. □

Theorem 6.2. *Let $1 \le k(n) \le n - 1$. Every sequence of values of n has a subsequence along which one of the following holds.*

(i) $k = \kappa$ *is constant, and* $(D - \mathbb{E}D)/\sqrt{\operatorname{Var} D}$ *converges in law to the standardization of* D_κ^∞.
(ii) $k \to \infty$ *and* $n - k \to \infty$, *and* $(D - \mathbb{E}D)/\sqrt{\operatorname{Var} D} \Rightarrow L_\infty$.
(iii) $n - 1 - k = r \geq 1$ *is constant, and* $(D - \mathbb{E}D)/\sqrt{\operatorname{Var} D} \Rightarrow L_r$.
(iv) $k = n - 1$; *then* $D = n - 1$ *and the standardized degree is not defined.*

If $n-1-k(n)$ *alternates between* 1 *and* 2, *the standardized degree has the two different subsequential limits* L_1 *and* L_2, *so it does not converge.*

Proof. Pass to a subsequence along which k is bounded or tends to infinity. If k is bounded, a further subsequence makes it constant, and (i) follows from Proposition 6.1, the variance of the limit being positive. If $k \to \infty$, either $n - k \to \infty$ along a further subsequence, which is (ii) by Theorem 2.1, or $n - k$ is bounded, and then a further subsequence makes $n - 1 - k$ constant, which is (iii) by Theorem 4.2 or (iv). The last statement follows from (iii) and Section 4. □

7. The mean at complementary budgets

Write $H_{n,k} = k - D_{n,k}$, and let $I_{n,r}$ be the number of vertices that prefer their edge to the root in $LK(n, r)$.

Proposition 7.1. *Let* $n \geq 3$, $1 \leq k \leq n - 2$ *and* $r = n - 1 - k$. *Then* $H_{n,k}$ *has the law of* $I_{n,r} - D_{n,r}$, *and* $\mathbb{E}H_{n,k} = \mathbb{E}H_{n,r}$.

Proof. Replace every score U by $U' = 1 - U$, which has the same law. A vertex prefers its k smallest U-scores exactly when it does not prefer them at budget r for the scores U'. Let P'_v be the set of neighbours that v prefers at budget r for U', let I' be the number of i with $0 \in P'_i$, and let $D' = \#\{i \in P'_0 : 0 \in P'_i\}$. The root degree at budget k for U is

$$D_{n,k} = \#\{i \notin P'_0 : 0 \notin P'_i\} = (n-1-I') - \#\{i \in P'_0 : 0 \notin P'_i\} = (n-1-I') - (r-D') = k - I' + D'.$$

So $H_{n,k} = I' - D'$, and (I', D') has the law of $(I_{n,r}, D_{n,r})$. Every vertex prefers r of its $n - 1$ neighbours, so by symmetry $\mathbb{E}I_{n,r} = r$, and $\mathbb{E}H_{n,k} = r - \mathbb{E}D_{n,r} = \mathbb{E}H_{n,r}$. □

Remark 7.2. The mean has a closed form. Let $f_{a,b}$ be the Beta(a, b) density. Since $B(k + 1, n - k - 1) = kB(k, n - k - 1)/(n - 1)$,

$$\frac{k}{x}\, f_{k+1,n-k-1}(x) = (n-1)\, f_{k,n-k-1}(x),$$

and (2.1) and (2.2) give, for $1 \leq k \leq n - 2$,

$$\mathbb{E}H_{n,k} = (n-1)\,\mathbb{E}(T' - T)_+ = (n-1)\int_0^1 F_T(t)\big(1 - F_T(t)\big)\, dt,$$

where T, T' are independent Beta$(k, n - 1 - k)$ variables. Since $1 - T \sim$ Beta$(n - 1 - k, k)$ and $F_{1-T}(t) = 1 - F_T(1 - t)$, the integral takes the same value at the budgets k and $n - 1 - k$, which is Proposition 7.1 for the means once more. For $k = 1$ the formula gives $\mathbb{E}H_{n,1} = (n - 2)/(2n - 3)$.

Corollary 7.3. *For every* $k \geq 1$ *the constant* c_k *of* [2, Theorem 7] *equals* μ_k; *both are* $(2k - 1)!/((k - 1)!^2\, 2^{2k-1})$.

Proof. Fix k and let $n \to \infty$. By [2, Theorem 7], $\mathbb{E}H_{n,k} \to c_k$. By Proposition 7.1 and Theorem 4.2 with complement k, $\mathbb{E}H_{n,k} = \mathbb{E}H_{n,n-1-k} \to \mu_k$. The common value follows from Proposition 4.3. □

The laws at complementary budgets are not the same. For $k = 1$, $H_{n,1}$ takes the values 0 and 1 and converges to a Bernoulli variable with mean $1/2$ and variance $1/4$, while $H_{n,n-2}$ converges to H_1, whose variance is $\nu_1 = 7/12$.

8. Computations

The program `checks.py` in the repository named under Data availability tests the statements above; its output, `checks_output.txt`, records 52 checks, all of which pass. Exact computations use rational arithmetic.

(a) For $n = 4$ and $n = 5$ the program goes through all $\binom{n}{2}!$ orders of the scores (720 and 3 628 800) and records the joint law of D and I for every budget. The mean and variance of H agree with Lemma 2.3, the identity $\mathbb{E}[p_X(W) \mid X] = q_n(X)$ of Lemma 2.2 holds as an identity of polynomials, and the laws in Proposition 7.1 agree. For instance $\mathbb{E}D_{5,2} = 52/35$ and $\operatorname{Var} D_{5,2} = 4889/11025$.

(b) The equality $\mathbb{E}H_{n,k} = \mathbb{E}H_{n,n-1-k}$ and the formula of Remark 7.2 hold exactly for all $n \le 24$ and $1 \le k \le n-2$, with $\mathbb{E}H_{n,k}$ computed from (2.1) and (2.2).

(c) In 200 000 samples each of $LK(12, 4)$ and $LK(12, 8)$, the sample mean and variance of D are within 0.2 standard errors of the exact values, which are 3.133127 and 0.914105 for $k = 4$ and 7.203892 and 0.987840 for $k = 8$.

(d) The sum (4.3) equals $r\binom{2r}{r}4^{-r}$ and c_r for $r \le 60$, and the values (4.5) hold. For $r = 1$ the exact pairs $(\mathbb{E}H, \operatorname{Var} H)$ at $n = 10, 20, 40, 80$ are $(0.470588, 0.523645)$, $(0.486486, 0.556124)$, $(0.493506, 0.570304)$ and $(0.496815, 0.576954)$, against $(\mu_1, \nu_1) = (0.5, 0.583333\ldots)$.

(e) Table 1 lists $\mathbb{E}H/s_n$ and $\operatorname{Var} H/s_n^2$ for three choices of k, computed from (2.2) and Lemma 2.3 by numerical integration. The integration agrees with the exact rational values at $(n, k) = (40, 20)$ and $(80, 30)$ to twelve decimal places. At $k = n/2$ the difference $\operatorname{Var} H/s_n^2 - v$ is close to $0.141/s_n$ for each n in the table.

k	n	$\mathbb{E}H/s_n$	$\operatorname{Var} H/s_n^2$
$n/2$	10^3	0.563695	0.299598
	10^4	0.564140	0.293509
	10^5	0.564185	0.291580
	10^6	0.564189	0.290970
$n/10$	10^3	0.563446	0.293487
	10^4	0.564115	0.291614
	10^5	0.564182	0.290984
	10^6	0.564189	0.290782
$\sqrt{n}$	10^2	0.556792	0.297426
	10^4	0.563484	0.290570
	10^6	0.564119	0.290654
	10^8	0.564183	0.290684
limit		$1/\sqrt{\pi} = 0.564190$	$v = 0.290688$

Table 1. The normalized mean and variance of $H = k - D$ in the interior regime.

The file `RootDegree.lean`, checked with Lean 4 [3] and Mathlib [5], proves the identity $\binom{n-3}{k-1}B(k, n-k-1) = 1/(n-2)$ used in Lemma 2.2, the identity $k\binom{2k}{k}4^{-k} = (2k-1)\binom{2k-2}{k-1}2^{1-2k}$ behind Corollary 7.3 for every $k \ge 1$, the agreement of (4.3) with $r\binom{2r}{r}4^{-r}$ for $r \le 6$, and the values of $M_{2,r}$, ν_r and μ_r^2/ν_r for $r = 1, 2$.

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics. Between 17 and 19 September 2026 agents running in OpenAI's Codex (GPT-5.6 Sol) obtained the results of Sections 2 to 7, and other Codex agents checked them. On 14 October 2026 Claude Opus 5.5 (Anthropic), working in Claude Code, checked the proofs line by line, wrote out in full the uniform integrability in Lemma 2.5 and the proofs of Theorem 4.2 and Proposition 6.1, and wrote the computer checks, the Lean file and the paper in its present form. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Data availability

No data were used. The code and the Lean file are available at
`https://github.com/yspennstate/bilateral-preference-degree-limits`.

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